Dallas International School

Test 6

Topics 4.5-4.6-4.11

Calculator Allowed

12th IB - Math AA

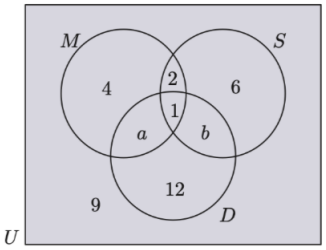
For SL & HL Students

Name : . . . . . .

EXERCISE 1 [3]

The Venn diagram opposite indicates the types of program a group of 40 individuals watched on television last night.

M represents movies, S represents sport, and D represents drama.



a Given that 10 people watched a movie last night, calculate a and b.

b Find the probability that one of these individuals, selected at random, watched:

i sport **ii** drama and sport **iii** a movie but not sport

iv drama but not a movie **v** drama or a movie.

EXERCISE 2

1. Find the probability of getting at least one head when a coin is flipped 4 times. [2]

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2. Find the smallest n such that P(at least one head in n flips of a coin) $> 99\%$. [3]

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EXERCISE 3 In a group of 40 boys, 23 have dark hair, 18 have brown eyes, and 26 have dark hair, brown eyes, or both. One of the boys is selected at random. Determine the probability that he has :

1. dark hair and brown eyes [1]

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2. neither dark hair nor brown eyes [1]

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3. dark hair but not brown eyes [1]

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4. brown eyes given that he has dark hair. [2]

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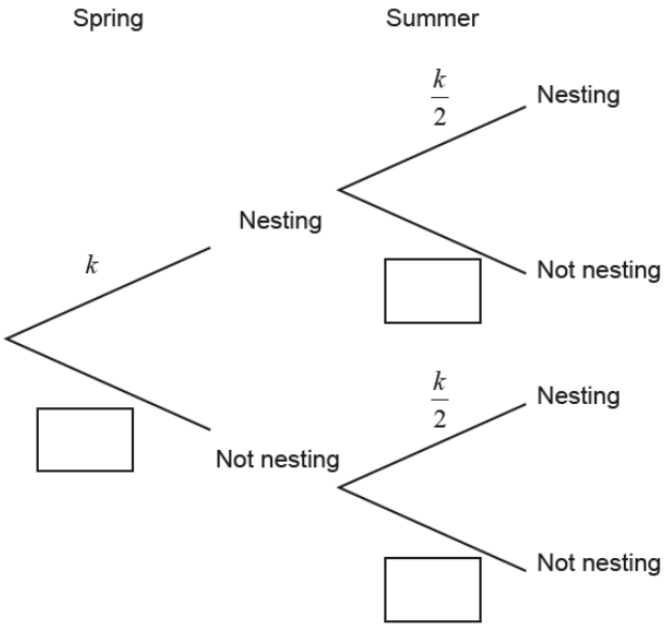
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EXERCISE 4 A species of bird can nest in two seasons : Spring and Summer.
The probability of nesting in Spring is k.

The probability of nesting in Summer is $\frac{k}{2}$.
This is shown in the following tree diagram.

1. Complete the tree diagram to show the probabilities of not nesting in each season. Write your answers in terms of k. [2]



It is known that the probability of not nesting in Spring and not nesting in Summer is $\frac{5}{9}$.

2. a. Show that $9k^2 - 27k + 8 = 0$. [3]

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b. Both $k = \frac{1}{3}$ and $k = \frac{8}{3}$ satisfy $9k^2 - 27k + 8 = 0$. [3]

State why $k = \frac{1}{3}$ is the only valid solution. [1]

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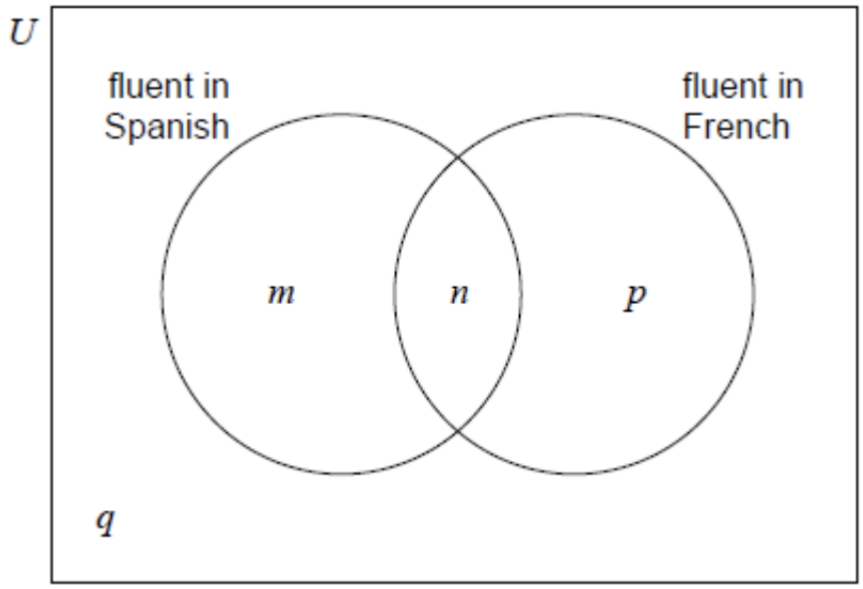
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EXERCISE 5 In a class of 30 students, 18 are fluent in Spanish, 10 are fluent in French, and 5 are not fluent in either of these languages. The following Venn diagram shows the events “fluent in Spanish” and “fluent in French”.

The values m, n, p and q represent numbers of students.



1. Write down the value of q. [1]

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2. Find the value of n. [2]

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3. Write down the value of m and of p. [3]

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Answer Key